Classical and Bayesian Estimation for Block Progressively Censored Competing Risks from the Kumaraswamy Distribution

Harsh Verma & Deepak Dhakad
Mehta Family School of Data Science and AI, IIT Guwahati, India

Abstract

We investigate failure times when an item can fail from either of two causes and some surviving items are removed during testing. We model the failure times with a Kumaraswamy distribution under block progressive Type-II censoring. The shape parameter is shared across blocks, while the cause parameters can vary. We develop maximum likelihood and Bayesian estimators, both with and without an order restriction within each block. Classical inference uses a profile likelihood; Bayesian inference uses either an independent scale prior or a hierarchical Gamma prior. We explain uncertainty intervals, posterior checks, and reliability quantities, and provide a reproducible simulation design and a real-data application. The exploratory simulation comprises 60 saved datasets and compares all six methods with explicit diagnostic screening. Both classical methods succeed throughout; Bayesian fits are available on smaller subsets. A published 39-failure appliance dataset illustrates the effect of imposing an ordering assumption that the recorded cause frequencies do not support.
Keywords: Kumaraswamy distribution; competing risks; block progressive censoring; maximum likelihood estimation; Bayesian estimation; order restriction; credible interval; simulation.

1 Introduction

In reliability studies, an item can fail for different reasons. We observe the time of the first failure and, when known, its cause. Sometimes testing ends before every item fails. Progressive Type-II censoring allows a predetermined number of surviving items to be removed after each observed failure. In block progressive censoring, this procedure is conducted separately in different groups or facilities.

The Kumaraswamy distribution is a flexible model for times measured on a fixed interval $(0, 1)$. Previous studies consider block progressive competing risks under another distribution [2], progressively censored Kumaraswamy competing risks [3], and Kumaraswamy competing risks with a different hybrid censoring scheme [4]. We combine the specified block censoring design with a shared-shape Kumaraswamy model. The common shape, independence assumptions and proposed order restriction are modeling choices; they require justification for a particular application.

The aim of this study is to see how the estimation methods behave when the two causes have different strengths, when their parameters are equal, and when an assumed order is incorrect. We first describe the data and likelihood, then derive the classical and Bayesian approaches. Finally, we explain the simulation design, what each performance measure tells us, and how we apply the model to observed appliance-failure data.

2 Model and Likelihood

2.1 The Kumaraswamy competing-risks model

Suppose there are k independent blocks. In block i, each item has two independent latent failure times

$$T_{iu1} \sim \mathrm{Kw}(a, b_{i1}), \qquad T_{iu2} \sim \mathrm{Kw}(a, b_{i2}), \qquad a, b_{i1}, b_{i2} > 0. \tag{1}$$

The shape parameter a is common to all blocks and causes. The parameters b_{i1}, b_{i2} may differ between blocks. Items are independent, and items within the same block have identical distributions. We observe $X_{iu} = \min(T_{iu1}, T_{iu2})$ and the cause $\delta_{iu} \in \{1, 2\}$ of that first failure. All recorded causes are assumed known.

For $T \sim \mathrm{Kw}(a, b)$ and $0 < t < 1$,

$$F(t) = 1 - (1 - t^a)^b, \qquad f(t) = ab\, t^{a-1}(1 - t^a)^{b-1}, \tag{2}$$

$$S(t) = (1 - t^a)^b, \qquad h(t) = \frac{ab\, t^{a-1}}{1 - t^a}. \tag{3}$$

Put $B_i = b_{i1} + b_{i2}$. Independence gives

$$S_i(t) = P(X_{iu} > t) = (1 - t^a)^{B_i}, \qquad X_{iu} \sim \mathrm{Kw}(a, B_i). \tag{4}$$

The joint density of the first failure time and cause c is

$$g_{ic}(t) = f(t; a, b_{ic}) S(t; a, b_{i,3-c}) = ab_{ic} t^{a-1}(1 - t^a)^{B_i - 1}. \tag{5}$$

Its integral is the probability of cause c:

$$P(\delta_{iu} = c) = \frac{b_{ic}}{B_i}. \tag{6}$$

Since $g_{ic}(t)$ is the minimum-lifetime density multiplied by this constant probability, cause and observed lifetime are independent under our model.

2.2 Block progressive censoring and likelihood

In block i, begin with n_i items. Observe m_i failures at ordered times $0 < x_{i1} < \cdots < x_{im_i} < 1$, recording each cause δ_{ij}. Immediately after failure j, randomly withdraw r_{ij} surviving items, independently of their future lifetimes and causes. The fixed withdrawal scheme satisfies

$$n_i = m_i + \sum_{j=1}^{m_i} r_{ij}. \tag{7}$$

The number at risk immediately before failure j is

$$Y_{ij} = n_i - (j - 1) - \sum_{\ell < j} r_{i\ell}. \tag{8}$$

An observed failure contributes the cause-specific density $g_{i,\delta_{ij}}(x_{ij})$. Each withdrawn survivor contributes $S_i(x_{ij})$. For the ordered failure times and recorded cause sequence, the block likelihood is

$$L_i = C_i \prod_{j=1}^{m_i} g_{i,\delta_{ij}}(x_{ij}) S_i(x_{ij})^{r_{ij}}, \qquad C_i = \prod_{j=1}^{m_i} Y_{ij}. \tag{9}$$

The constant C_i accounts for the possible items that could fail at each stage; it does not depend on the unknown parameters. The blocks are independent, so $L = \prod_i L_i$.

Let d_{ic} be the number of observed failures from cause c in block i, $M = \sum_i m_i$, and define

$$T = \sum_{i,j} \log x_{ij}, \qquad H_i(a) = \sum_j \log(1 - x_{ij}^a), \tag{10}$$

$$W_i(a) = -\sum_j (r_{ij} + 1) \log(1 - x_{ij}^a), \qquad C = \prod_i C_i. \tag{11}$$

Substituting (4) and (5) into (9), then taking logarithms, gives the full log-likelihood

$$\boxed{\ell = \log C + M \log a + (a-1)T + \sum_{i,c} d_{ic} \log b_{ic} - \sum_i B_i W_i(a) - \sum_i H_i(a).} \tag{12}$$

Terms with a zero count are interpreted through the continuous convention $0 \log b = 0$.

3 Maximum Likelihood Estimation

3.1 Unrestricted maximum likelihood estimation

To maximize (12), differentiate with respect to each b_{ic}:

$$\frac{\partial \ell}{\partial b_{ic}} = \frac{d_{ic}}{b_{ic}} - W_i(a). \tag{13}$$

When $d_{ic} > 0$, setting this score to zero yields

$$\boxed{\widehat{b}_{ic}(a) = \frac{d_{ic}}{W_i(a)}, \qquad \widehat{B}_i(a) = \frac{m_i}{W_i(a)}.} \tag{14}$$

If $d_{ic} = 0$, the supremum occurs as $b_{ic} \downarrow 0$: this is a boundary limit, not an attained maximum under the strict positivity assumption.

Substitute (14) into (12). The resulting profile log-likelihood, up to terms constant in a, is

$$\ell_p(a) = M \log a + (a-1)T - \sum_i m_i \log W_i(a) - \sum_i H_i(a) + \text{constant}. \tag{15}$$

Its score is

$$U_p(a) = \frac{M}{a} + T - \sum_i m_i \frac{W_i'(a)}{W_i(a)} - \sum_i H_i'(a). \tag{16}$$

We solve $U_p(a) = 0$ numerically with a bracketed root method, and substitute $\widehat{a}$ into (14). The profile is strictly concave in a and has a unique finite maximizer if at least one block has two or more distinct observed failure times. If every block has only one observed failure, the profile likelihood is unbounded as $a \to \infty$. A finite positive-parameter maximum also requires both cause counts to be positive in each block. These conclusions concern the mathematical likelihood; numerical failure of an optimizer is a separate issue.

3.2 Maximum likelihood estimation with an order restriction

Assume, for each block separately,

$$b_{i1} \le b_{i2}. \tag{17}$$

This is an additional modeling assumption, not an order imposed between blocks. Set

$$B_i = b_{i1} + b_{i2}, \qquad p_i = \frac{b_{i1}}{B_i}. \tag{18}$$

The restriction is simply $p_i \le 1/2$, and the log-likelihood separates into a part involving (a, B_i) and a cause-count part

$$\ell = \ell_{\text{time}}(a, \mathbf{B}) + \sum_i \{d_{i1} \log p_i + d_{i2} \log(1 - p_i)\}. \tag{19}$$

The cause-count part is maximized at d_{i1}/m_i without a restriction. Under (17), its maximum over $0 \le p_i \le 1/2$ is therefore

$$\boxed{\widetilde{p}_i = \min\left(\frac{d_{i1}}{m_i}, \frac{1}{2}\right), \qquad \widetilde{b}_{i1} = \widehat{B}_i \widetilde{p}_i, \quad \widetilde{b}_{i2} = \widehat{B}_i(1 - \widetilde{p}_i).} \tag{20}$$

Thus we keep the unrestricted pair if it satisfies the order; otherwise both parameters are pooled at $\widehat{B}_i/2$. The optimized cause-count term depends only on counts, not on a or B_i. Consequently,

$$\boxed{\widetilde{a} = \widehat{a}, \qquad \widetilde{B}_i = \widehat{B}_i.} \tag{21}$$

In particular, the restricted and unrestricted fitted block survival and hazard functions are identical. A strict violation in observed cause proportions causes a loss of maximized log-likelihood under the restriction. If $d_{i1} = 0$, the restricted solution also lies on a zero boundary; if $d_{i2} = 0$, pooling yields two positive equal parameters. The equality constraint is permitted, but a zero parameter is not in the original strictly positive space.

3.3 Observed information and confidence intervals

The observed information is the negative Hessian of the log-likelihood. To simplify its expression, use the parameter vector $q = (a, B_1, \ldots, B_k, p_1, \ldots, p_k)$. Put $w_{ij} = r_{ij} + 1$, $u_{ij} = -\log x_{ij}$ and

$$P_{ij} = \frac{u_{ij}}{e^{a u_{ij}} - 1}, \qquad Q_{ij} = \frac{u_{ij}^2 e^{-a u_{ij}}}{(1 - e^{-a u_{ij}})^2}. \tag{22}$$

Differentiating (19), the nonzero entries of $J(q) = -\nabla^2 \ell(q)$ are

$$J_{aa} = \frac{M}{a^2} + \sum_{i,j} (B_i w_{ij} - 1) Q_{ij}, \qquad J_{aB_i} = -\sum_j w_{ij} P_{ij}, \tag{23}$$

$$J_{B_i B_i} = \frac{m_i}{B_i^2}, \qquad J_{p_i p_i} = \frac{d_{i1}}{p_i^2} + \frac{d_{i2}}{(1 - p_i)^2}. \tag{24}$$

All other entries vanish, except the symmetric counterparts. For a regular unrestricted interior MLE, $J(\widehat{q})^{-1}$ estimates its covariance matrix, and the square roots of its diagonal entries give standard errors. For a positive parameter θ, an approximate 95% log-scale interval is

$$\left[\widehat{\theta} \exp\left(-1.96 \frac{\mathrm{SE}(\widehat{\theta})}{\widehat{\theta}}\right), \ \widehat{\theta} \exp\left(1.96 \frac{\mathrm{SE}(\widehat{\theta})}{\widehat{\theta}}\right)\right]. \tag{25}$$

The covariance for the original b parameters follows from the Jacobian of $b_{i1} = B_i p_i$ and $b_{i2} = B_i(1 - p_i)$. These intervals are large-sample approximations, not exact guarantees.

Under our model the observed cause count satisfies $d_{i1} \sim \text{Binomial}(m_i, p_i)$, conditional on the fixed number of failures. Thus an exact Clopper–Pearson confidence interval can be formed for each p_i. Under the order restriction, intersect this interval with $[0, 1/2]$. If the intersection is empty, report it as empty; do not force it to contain the restricted estimate. Ordinary unrestricted Wald intervals are not automatically valid for cause parameters at active constraints. The currently developed methods do not assign such intervals to restricted individual b_{ic} parameters.

3.4 Reliability estimation

Using $\widehat{a}$ and $\widehat{B}_i$, the fitted block survival, hazard and median time to failure are

$$\widehat{S}_i(t) = (1 - t^{\widehat{a}})^{\widehat{B}_i}, \tag{26}$$

$$\widehat{h}_i(t) = \frac{\widehat{a}\widehat{B}_i t^{\widehat{a}-1}}{1 - t^{\widehat{a}}}, \tag{27}$$

$$\widehat{v}_i = \left(1 - 2^{-1/\widehat{B}_i}\right)^{1/\widehat{a}}. \tag{28}$$

The fitted cause-c cumulative incidence is

$$\widehat{G}_{ic}(t) = \widehat{p}_{ic}\{1 - \widehat{S}_i(t)\}, \qquad \widehat{p}_{ic} = \frac{\widehat{b}_{ic}}{\widehat{B}_i}. \tag{29}$$

For regular estimates, approximate standard errors of these functions can be obtained by the delta method: if $g(q)$ is a reliability quantity, $\widehat{\mathrm{Var}}\{g(\widehat{q})\} = \nabla g(\widehat{q})^{\mathsf{T}} J(\widehat{q})^{-1} \nabla g(\widehat{q})$. Overall survival, hazard and median use only (a, B_i) and are unchanged by the within-block restriction. Cause-specific incidence also depends on p_i and may change.

4 Bayesian Estimation

We use the same likelihood in (12). Throughout, Gamma distributions are parameterized by *shape* and *rate*. The squared-error Bayes estimate is a posterior mean, provided that mean exists. We study two prior structures, each with and without the within-block order restriction (17). The notation $\pi(\cdot \mid D)$ denotes a posterior density, up to proportionality where normalization is not written.

4.1 Independent scale prior without ordering

Consider the improper independent scale prior

$$\pi_I(a, \mathbf{b}) \propto a^{-1} \prod_{i=1}^{k} \prod_{c=1}^{2} b_{ic}^{-1}, \qquad a, b_{ic} > 0. \tag{30}$$

This is a scale-invariant prior specification, *not* a claim that it is the joint Jeffreys prior. Combining (30) with (12), the posterior kernel is

$$\pi_I(a, \mathbf{b} \mid D) \propto a^{M-1} e^{(a-1)T - \sum_i H_i(a)} \prod_{i,c} b_{ic}^{d_{ic}-1} e^{-b_{ic} W_i(a)}. \tag{31}$$

For $d_{ic} > 0$, the full conditionals are independent Gamma distributions:

$$b_{ic} \mid a, D \sim \mathrm{Gamma}(d_{ic}, W_i(a)). \tag{32}$$

Integrating out all cause parameters gives a one-dimensional marginal kernel for a:

$$\boxed{h_I(a) \propto a^{M-1} \exp\left\{(a-1)T - \sum_i H_i(a)\right\} \prod_i W_i(a)^{-m_i}.} \tag{33}$$

This density can be sampled using a log-scale slice update for $\eta = \log a$, including the transformation Jacobian e^{η}. The b_{ic} can then be drawn conditionally from (32).

The improper prior requires explicit posterior propriety checks. Set $y_{ij} = -\log x_{ij}$, $y_i^* = \min_j y_{ij}$, and $D_* = \sum_{i,j}(y_{ij} - y_i^*)$. For the unrestricted independent-scale analysis, the posterior is proper when every $d_{ic} > 0$, $M > 1$ and $D_* > 0$; outside that domain the implementation does not return posterior estimates. Moreover,

$$E_I(b_{ic}^r \mid D) < \infty \quad \Longleftrightarrow \quad D_* > r y_i^*, \qquad r > 0. \tag{34}$$

A proper posterior does not by itself guarantee a finite posterior mean or variance. The implementation checks moment existence separately rather than relying on finite sampled averages.

4.2 Independent scale prior with within-block ordering

For each block use $B_i = b_{i1} + b_{i2}$ and $p_i = b_{i1}/B_i$. The Jacobian is B_i. Transforming the joint posterior in (31) gives the joint kernel

$$\pi_I(a, \mathbf{B}, \mathbf{p} \mid D) \propto a^{M-1} e^{(a-1)T - \sum_i H_i(a)} \prod_i \left\{ B_i^{m_i - 1} e^{-B_i W_i(a)} p_i^{d_{i1}-1} (1-p_i)^{d_{i2}-1} \right\}. \tag{35}$$

Integrating over B_i and p_i recovers the marginal density $h_I(a)$ in (33). In particular,

$$B_i \mid a, D \sim \text{Gamma}(m_i, W_i(a)), \qquad p_i \mid D \sim \text{Beta}(d_{i1}, d_{i2}). \tag{36}$$

The restriction $b_{i1} \le b_{i2}$ is equivalent to $p_i \le 1/2$. Under the restricted posterior, only p_i is truncated to $(0, 1/2]$; $h_I(a)$ and the conditional distribution of each B_i are unchanged. Thus the full posterior distributions of overall block survival, hazard and median are invariant to ordering *for this independent prior*. Cause proportions and cause-specific incidence need not be invariant. Restricted posterior means are calculated from constrained draws, rather than by projecting or pooling unrestricted posterior means.

The implementation conservatively requires both cause counts positive even under restriction. This is a computational support rule, not a mathematical assertion that all restricted zero-count posteriors are improper: some cases with $d_{i2} = 0 < d_{i1}$ can be proper after conditioning on $p_i \le 1/2$.

4.3 Hierarchical Gamma prior without ordering

We also consider a proper hierarchical model in which cause-specific parameters share information across blocks:

$$a \sim \text{Gamma}(A_a, B_a), \tag{37}$$

$$b_{ic} \mid \kappa_c, \lambda_c \overset{\text{ind}}{\sim} \text{Gamma}(\kappa_c, \lambda_c), \tag{38}$$

$$\kappa_c \sim \text{Gamma}(A_c, B_c), \qquad \lambda_c \sim \text{Gamma}(C_c, D_c), \qquad c = 1, 2. \tag{39}$$

All fixed hyperparameters are positive and the two causes have separate hyperparameter pairs. Up to a factor independent of unknowns, the joint posterior is

$$\pi_H(a, \mathbf{b}, \boldsymbol{\kappa}, \boldsymbol{\lambda} \mid D) \propto a^{M + A_a - 1} e^{-B_a a + (a-1)T - \sum_i H_i(a)} \times \prod_{c=1}^{2} \left[\frac{\kappa_c^{A_c - 1} e^{-B_c \kappa_c} \lambda_c^{k\kappa_c + C_c - 1} e^{-D_c \lambda_c}}{\Gamma(\kappa_c)^k} \prod_i b_{ic}^{\kappa_c + d_{ic} - 1} e^{-(\lambda_c + W_i(a)) b_{ic}} \right]. \tag{40}$$

In particular, neither $\lambda_c^{k\kappa_c}$ nor $\Gamma(\kappa_c)^{-k}$ may be omitted: both depend on unknown hyperparameters. The conjugate full conditionals are

$$b_{ic} \mid \text{rest}, D \sim \text{Gamma}(\kappa_c + d_{ic}, \lambda_c + W_i(a)), \tag{41}$$

$$\lambda_c \mid \text{rest}, D \sim \text{Gamma}\left(k\kappa_c + C_c, D_c + \sum_i b_{ic}\right). \tag{42}$$

The nonconjugate conditionals for a and κ_c are updated by slice sampling on the log scale. For reference,

$$\log \pi(\kappa_c \mid \text{rest}, D) = (A_c - 1)\log \kappa_c - B_c\kappa_c - k \log \Gamma(\kappa_c) + \kappa_c \left(k \log \lambda_c + \sum_i \log b_{ic}\right) + \text{const}, \tag{43}$$

$$\log \pi(a \mid \text{rest}, D) = (M + A_a - 1)\log a - B_a a + (a-1)T - \sum_i H_i(a) - \sum_i B_i W_i(a) + \text{const}. \tag{44}$$

The change of variables in each log-scale slice update includes its Jacobian. For valid progressively censored data, positive fixed Gamma hyperparameters lead to a proper posterior. The code distinguishes verified finite moments from moments that are not established by its sufficient bounds.

4.4 Hierarchical Gamma prior under within-block ordering

The restricted hierarchical model conditions the *whole joint posterior* in (40) on the event

$$\mathcal{C} = \bigcap_{i=1}^{k} \{b_{i1} \le b_{i2}\}. \tag{45}$$

The Gibbs update for b_{i1} is its Gamma conditional truncated above at the current b_{i2}; the update for b_{i2} is its Gamma conditional truncated below at the newly sampled b_{i1}. The remaining full conditionals retain their forms because the event indicator is then fixed. We do not add separate hyperparameter-dependent normalizing factors corresponding to an independently truncated conditional prior; doing so would define a different model.

Unlike the independent scale-prior construction, the restriction here can alter the marginal posterior of a, the totals B_i and hence overall reliability. No across-block ordering is imposed.

4.5 Bayesian estimates, intervals, diagnostics and limitations

All four Bayesian analyses use posterior means under squared-error loss where the requisite first moments are established. For any fixed $0 < t < 1$, the posterior reliability functions are calculated for each joint draw: $S_i(t)$, $h_i(t)$, the median v_i, the cause probabilities p_i and the cause-specific cumulative incidence $G_{ic}(t)$. Averaging draw-wise functions is generally different from evaluating a nonlinear function at posterior mean parameters.

Marginal intervals are the shortest contiguous intervals containing at least a prespecified fraction (typically 0.95) of pooled retained draws. These approximate highest-posterior-density (HPD) intervals for suitable unimodal marginals; they are not guaranteed to represent disconnected HPD regions and are not invariant under reparameterization. Credible probabilities are posterior probabilities, not empirical frequentist coverage claims.

Sampling uses multiple independently initialized chains, a warm-up phase, and retained draws. Diagnostics include rank-normalized and folded split $\widehat{R}$, bulk and tail effective sample sizes, and

Monte Carlo standard errors when posterior second moments are established. The configured default checks are $\widehat{R} \leq 1.01$ and bulk and tail ESS ≥ 400; these are diagnostics rather than proofs of convergence. Numerical failures, convergence failures, posterior impropriety and unsupported moment conditions are recorded explicitly. The sensitivity of hierarchical estimates to prior choices remains a limitation. In the numerical study below, all reported Bayesian point estimates and intervals are restricted to fits that passed the recorded diagnostic and moment checks.

5 Numerical Illustration

5.1 Simulation Study

The simulation uses $k = 2$ independent blocks and two latent failure causes per item. For each unit independently draw $U_{iuc} \sim \text{Uniform}(0, 1)$ and set

$$T_{iuc} = \left[1 - (1 - U_{iuc})^{1/b_{ic}}\right]^{1/a}, \qquad X_{iu} = \min(T_{iu1}, T_{iu2}), \quad \delta_{iu} = \arg\min_{c=1,2} T_{iuc}. \tag{46}$$

At every observed failure the item with the earliest remaining lifetime fails, after which the prescribed number r_{ij} of surviving items is withdrawn uniformly at random. Merely keeping the first m_i complete-data order statistics would not implement progressive censoring correctly. Every observed dataset is saved with scenario, seed, times, causes, withdrawals and true parameters; all six estimation procedures receive that same observed dataset.

The submission-scale experiment uses $a = 1.5$, $n_1 = n_2 = 50$, $m_1 = m_2 = 30$, and $r_{i,m_i} = 20$ with all earlier removals zero. Three principal scenarios are:

True ordering	(b_{11}, b_{12})	(b_{21}, b_{22})
Strict	$(0.5, 1.5)$	$(0.75, 2.25)$
Equality	$(1, 1)$	$(1.5, 1.5)$
Violated	$(1.5, 0.5)$	$(2.25, 0.75)$

All scenarios preserve $B_1 = 2$ and $B_2 = 3$, isolating cause allocation while keeping the marginal minimum-lifetime distributions fixed. There are 20 generated datasets in each scenario. This makes the study useful for illustrating patterns, but not for drawing high-precision conclusions about coverage or differences in MSE. Other sample sizes and censoring plans would be useful in a larger study.

For each scenario, method m and parameter θ, let $\mathcal{V}_{m\theta}$ denote the repetitions with a valid reported estimate and $N_{m\theta} = |\mathcal{V}_{m\theta}|$. We compute

$$\overline{\widehat{\theta}}_m = \frac{1}{N_{m\theta}} \sum_{r \in \mathcal{V}_{m\theta}} \widehat{\theta}_m^{(r)}, \tag{47}$$

$$\text{AB}_m(\theta) = \left|\overline{\widehat{\theta}}_m - \theta_0\right|, \qquad \text{MSE}_m(\theta) = \frac{1}{N_{m\theta}} \sum_{r \in \mathcal{V}_{m\theta}} \left(\widehat{\theta}_m^{(r)} - \theta_0\right)^2.$$

For a supported interval $[L_m^{(r)}, U_m^{(r)}]$, use the valid interval subset $\mathcal{I}_{m\theta}$ with size $N_{m\theta}^I$ and calculate

$$\text{CP}_m(\theta) = \frac{1}{N_{m\theta}^I} \sum_{r \in \mathcal{I}_{m\theta}} \mathbf{1}\{L_m^{(r)} \leq \theta_0 \leq U_m^{(r)}\}, \qquad \text{AW}_m(\theta) = \frac{1}{N_{m\theta}^I} \sum_{r \in \mathcal{I}_{m\theta}} (U_m^{(r)} - L_m^{(r)}). \tag{48}$$

Every summary reports its valid numerator and denominator as well as method-specific numerical and MCMC failure rates. Empty confidence sets, where defined, have zero coverage and zero set length; unavailable individual constrained-parameter intervals are marked not applicable, never fabricated. Restricted-model results under violated true ordering are a deliberate misspecification assessment. A method's smaller conditional MSE should not conceal a poor success rate.

How to read the numerical results

Absolute bias (AB) measures systematic departure from the true parameter: smaller values are preferable. Mean squared error (MSE) also penalizes variability, so a method can have small bias but still have a large MSE. Coverage probability (CP) checks how often an interval contains the true parameter; we compare it with the nominal 95% target. Average width (AW) measures interval precision, but a narrow interval is only attractive when its coverage is adequate. Because the simulation is small and Bayesian fits may not complete for every dataset, we give the number of usable replications beside every metric. For fair head-to-head MSE comparisons, we use the same successfully analyzed datasets for both methods.

Execution and usable sample sizes

All 60 simulated datasets were generated and saved before fitting. Both unrestricted and restricted MLEs succeeded on all 60 datasets (20 per scenario). Of 240 planned Bayesian fits, 120 passed diagnostic checks: 31 independent-prior unrestricted, 31 independent-prior restricted, 31 hierarchical unrestricted, and 27 hierarchical restricted. The restricted hierarchical method had 11 additional diagnostic failures, and the remaining Bayesian tasks were not run within the computational budget. Thus complete six-method comparisons were available on 12 strict-order datasets, 13 equality datasets, and only one order-violation dataset. Failed and unattempted fits are not treated as successful observations. Results below are descriptive and exploratory rather than precision estimates of estimator ranking.

Table 1: Successful fits in the simulation. Each method was planned for 20 datasets per scenario. The last four methods use posterior means only when convergence and moment checks permit them.

Method	Strict	Equality	Violation	Total
Unrestricted MLE	20	20	20	60
Restricted MLE	20	20	20	60
Independent Bayes (U)	12	14	5	31
Independent Bayes (R)	12	14	5	31
Hierarchical Bayes (U)	12	13	6	31
Hierarchical Bayes (R)	12	14	1	27
All six jointly available	12	13	1	26

Parameter accuracy

Table 2 shows the observed MSE for the common shape parameter and the two competing-risk coefficients in block A. These selected quantities illustrate both total-lifetime and cause-specific accuracy. The corresponding complete numerical exports cover all five primary parameters and additional reliability quantities. Values calculated from different numbers of valid fits must not be interpreted as matched head-to-head comparisons; the repository also contains comparisons restricted to identical saved datasets.

Strict ordering. The classical restricted and unrestricted results coincide in all 20 strict-order datasets: the empirical cause counts did not activate pooling. The independent and hierarchical posterior restrictions change the Bayesian cause estimates slightly. On the selected block-A coefficient b_{11}, restriction reduced observed Bayesian MSE from .0695 to .0612 with the independent prior and from .0688 to .0646 with the hierarchical prior, on their corresponding 12-run comparisons. It did not improve every parameter: for b_{12} the changes were small and slightly unfavorable.

Table 2: Observed MSE for the common shape a and block-A cause parameters, with valid repetition count R. The true values are $a = 1.5$ and scenario-specific b_{11}, b_{12}. Smaller MSE is better only within a comparable sample.

Scenario	Method	R	MSE(a)	MSE(b_{11})	MSE(b_{12})
Strict	MLE (U)	20	.0515	.0457	.3700
	MLE (R)	20	.0515	.0457	.3700
	Independent Bayes (U)	12	.0267	.0695	.1566
	Independent Bayes (R)	12	.0268	.0612	.1575
	Hierarchical Bayes (U)	12	.0187	.0688	.1311
	Hierarchical Bayes (R)	12	.0200	.0646	.1344
Equality	MLE (U)	20	.0377	.1265	.1830
	MLE (R)	20	.0377	.0963	.1904
	Independent Bayes (U)	14	.0426	.0394	.2201
	Independent Bayes (R)	14	.0426	.0562	.2167
	Hierarchical Bayes (U)	13	.0358	.0335	.2061
	Hierarchical Bayes (R)	14	.0329	.0432	.1966
Violation	MLE (U)	20	.0694	1.0008	.0641
	MLE (R)	20	.0694	.3962	.8296
	Independent Bayes (U)	5	.0257	1.7849	.0972
	Independent Bayes (R)	5	.0256	.3983	1.5956
	Hierarchical Bayes (U)	6	.0644	1.8206	.0989
	Hierarchical Bayes (R)	1	.0730	.4872	.1464

The hierarchical posterior shape MSE was smaller than the classical shape MSE in these saved summaries, but the sample sets differ, so this is not proof of superiority.

Equality. When the two cause parameters are equal in truth, pooling can reduce noise in their estimates. For block A, classical MSE for b_{11} fell from .1265 to .0963 with restriction, but MSE for b_{12} rose slightly from .1830 to .1904. The restricted Bayesian results are similarly mixed. Restriction therefore does not guarantee an improvement for every cause parameter, even when it is correctly specified.

Ordering violation. Here the restriction is deliberately false. The restricted classical model must enforce $b_{i1} \leq b_{i2}$, although the true coefficients satisfy the reverse inequality. In block A, classical MSE for b_{11} decreased from 1.0008 to .3962, whereas MSE for b_{12} rose sharply from .0641 to .8296. The apparent improvement in one coefficient reflects bias–variance tradeoffs and cannot be read as evidence that the false order assumption is correct. Independent Bayesian results show the same direction of tradeoff for these coefficients. Only one dataset has a successful restricted hierarchical fit jointly with all other methods, making any hierarchical comparison in this scenario especially fragile.

Interval coverage and precision

Table 3 summarizes the empirical coverage probability (CP) and average width (AW) for intervals for the common shape parameter a. These are marginal 95% intervals, with classical approximate confidence intervals and Bayesian shortest contiguous posterior-draw credible intervals treated separately. The classical shape interval has the same coverage for both MLE methods because the fitted shape parameter is invariant to the within-block restriction. The small denominators imply coarse coverage percentages, especially for Bayesian methods.

A recorded 100% CP with only five or twelve intervals does not establish perfect coverage. Similarly, a narrower Bayesian interval should not be declared superior without enough matched observations and reliable coverage. Classical restricted individual-b confidence intervals at active constraints are not presented as ordinary Wald intervals. The supplementary saved exports give separate interval denominators for every supported quantity.

Table 3: Empirical interval coverage and average width for a (nominal level 95%). R_I is the number of usable intervals.

Scenario	Method	R_I	CP	AW
Strict	MLE (U/R)	20	.950	.861
	Independent Bayes (U)	12	1.000	.843
	Independent Bayes (R)	12	1.000	.844
	Hierarchical Bayes (U)	12	1.000	.786
	Hierarchical Bayes (R)	12	1.000	.792
Equality	MLE (U/R)	20	1.000	.823
	Independent Bayes (U)	14	.929	.798
	Independent Bayes (R)	14	1.000	.803
	Hierarchical Bayes (U)	13	1.000	.744
	Hierarchical Bayes (R)	14	1.000	.758
Violation	MLE (U/R)	20	.900	.854
	Independent Bayes (U)	5	1.000	.796
	Independent Bayes (R)	5	1.000	.791
	Hierarchical Bayes (U)	6	.833	.769
	Hierarchical Bayes (R)	1	1.000	.617

Overall interpretation and limitations

The results lead to three clear findings within the tested design. First, classical estimation was computationally dependable: every planned MLE fit succeeded. Second, a within-block restriction changes how the total failure risk is divided between causes, not the classical fitted total lifetime distribution. The restriction does not uniformly lower MSE, even when it agrees with the true order. Third, Bayesian estimation gave competitive accuracy for selected parameters but was markedly more expensive, and the available results are insufficient to rank it above MLE overall. In particular, the restricted hierarchy had more diagnostic problems when the imposed order was false. Only the three late-withdrawal regimes at one sample size were run, so this is an illustrative, not exhaustive, simulation study. We do not claim a universal best method or general nominal-coverage validity from these runs.

5.2 Application to observed appliance failures

We also apply the methods to a published example involving 39 appliance failures. The observations are attributed to Nelson's reliability data and reproduced as Table 6 in a later study by Shi, Yan and Peng [1]. Each observation has a recorded failure time and one of two cause labels: failure mechanism 5 or all other recorded failure mechanisms. There are 26 failures in the first category and 13 in the second.

The source records no withdrawals or unobserved failure causes. We therefore use the *complete-data special case* of our likelihood, with one block, $n = m = 39$ and $r_j = 0$ for all observations. This checks whether the estimation procedures can be applied to genuine failures; it is not a real experiment with progressive removals. We use a single block rather than inventing an additional grouping.

The source reports times in thousands of cycles. For the Kumaraswamy support $(0, 1)$, we divide each value by 5, or equivalently divide the number of cycles by 5,000. The maximum reported lifetime is 4,129 cycles, so the transformed times lie inside the required interval. The 5,000-cycle limit is a chosen analysis scale, *not* a verified physical upper bound. Results should therefore be interpreted relative to this scale.

The intervals for a overlap substantially across methods. Under the unrestricted analyses, the intervals for b_1 and b_2 also overlap, so the observed difference in cause frequencies should not be presented as a precise separation between underlying risk parameters. The restriction changes the interpretation of the cause parameters more clearly than it changes the common shape estimate. Importantly, the reported Bayesian intervals summarize posterior uncertainty under each specified

Table 4: Estimates from the appliance-failure data, verified from the saved computational outputs. Bayesian entries are posterior means.

Method	a	b_1	b_2
Unrestricted MLE	1.0249	1.2567	0.6284
Restricted MLE	1.0249	0.9425	0.9425
Independent Bayes (U)	1.0164	1.2529	0.6244
Independent Bayes (R)	1.0113	0.8835	0.9917
Hierarchical Bayes (U)	1.0340	1.2835	0.6629
Hierarchical Bayes (R)	1.0310	0.9140	1.0262

Table 5: Marginal 95% uncertainty intervals for the appliance data. Classical entries use log-scale approximate confidence intervals; Bayesian entries use shortest contiguous posterior-sample credible intervals. A dash means that the relevant constrained-parameter interval is not supported by the classical procedure.

Method	Interval for a	Interval for b_1	Interval for b_2
MLE (unrestricted)	[0.718, 1.464]	[0.748, 2.110]	[0.330, 1.198]
MLE (restricted)	[0.718, 1.464]	—	—
Independent Bayes (U)	[0.652, 1.376]	[0.667, 1.936]	[0.261, 1.026]
Independent Bayes (R)	[0.651, 1.368]	[0.490, 1.325]	[0.545, 1.474]
Hierarchical Bayes (U)	[0.689, 1.375]	[0.691, 1.931]	[0.293, 1.077]
Hierarchical Bayes (R)	[0.680, 1.384]	[0.494, 1.354]	[0.554, 1.523]

prior and model; they are not confidence intervals obtained by inverting repeated sampling tests.

All six estimates were obtained successfully, and the four Bayesian fits passed the recorded diagnostics. The unrestricted classical estimate puts the first cause parameter above the second, matching the larger observed count for mechanism 5. Imposing $b_1 \leq b_2$ forces the classical estimate to $b_1 = b_2$, so it does not fit the observed cause split as closely. The unrestricted log-likelihood is 87.799 compared with 85.590 under the restriction. The two classical fits share the same a and $B = b_1 + b_2$, hence the same overall fitted survival curve, as predicted by the theory.

The recorded descriptive maximum difference between the fitted and empirical distribution functions is approximately 0.090; this is not a calibrated goodness-of-fit p-value. The source also discusses possible dependence between causes, whereas our latent failure times are assumed independent. Thus this application shows how the methods behave on real observations, but does not prove that the model assumptions hold. We do not compute simulation bias, MSE, or coverage for this dataset because the true parameters are unknown. Table 5 presents the primary marginal uncertainty intervals. Additional reliability predictions and computational diagnostics were recorded during the analysis. The application remains conditional on the selected scaling and independence assumptions.

6 Conclusion

This study provides a common framework for classical and Bayesian estimation with two competing causes and block progressive Type-II censoring. The classical problem becomes simpler after separating total failure risk from the proportions of the two causes. Under the within-block order restriction, the classical total risk and shape estimates stay the same, while cause parameters may be pooled. The independent-scale Bayesian model has a related invariance for overall lifetime quantities, whereas the hierarchical Gamma model does not generally have it.

The appliance example provides a direct practical conclusion: 26 of the 39 observed failures belonged to cause 1, yielding an unrestricted estimated cause-1 probability of 0.667. The stipulated restriction requires this probability to be at most 0.5; the restricted MLE therefore pools the cause

parameters and loses 2.209 units of maximized log-likelihood relative to the unrestricted fit. This makes the restriction unsuitable as an unquestioned assumption for this example, although the empirical cause split alone is not a calibrated model-selection test. The simulation compares this trade-off when the order is correct, at equality, and incorrect. Both MLE procedures succeeded on all 60 saved datasets, while 120 of the 240 planned Bayesian fits passed diagnostic checks. The restriction lowered MSE for some cause coefficients and increased it for others, notably under the deliberately false ordering assumption. Only 26 datasets allowed full six-method paired comparisons (12 strict, 13 equality, and one violation). Our definite conclusion is therefore conditional: both classical MLE procedures were efficient and numerically reliable in this experiment; the order constraint should be imposed only with scientific justification; and Bayesian methods offered useful posterior uncertainty quantification but no demonstrated universal accuracy advantage. The simulation is exploratory, especially with only one complete six-method comparison under order violation. More replications, alternate withdrawal plans, a real progressively censored dataset, and sensitivity checks for priors and time scaling are required for broader claims.

References

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